BRITISH STANDARD

BS EN 14792:2005

Stationary source emissions — Determination of mass concentration of nitrogen oxides (NOx) — Reference method: Chemiluminescence

The European Standard EN 14792:2005 has the status of a British Standard

ICS 13.040.40

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British Standards

National foreword

This British Standard is the official English language version of EN 14792:2005.

The UK participation in its preparation was entrusted by Technical Committee EH/2, Air quality, to Subcommittee EH/2/1, Stationary source emissions, which has the responsibility to:

— aid enquirers to understand the text;

— present to the responsible international/European committee any enquiries on the interpretation, or proposals for change, and keep UK interests informed;

— monitor related international and European developments and promulgate them in the UK.

A list of organizations represented on this subcommittee can be obtained on request to its secretary.

Cross-references

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EUROPEAN STANDARD

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EN 14792

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English Version

Stationary source emissions - Determination of mass concentration of nitrogen oxides (NOx) - Reference method: Chemiluminescence

missions de sources fixes - Détermination de la concentration massique en oxides d'azote (NOx) - Méthode de référence: Chimuluminescence

Emissionen aus stationären Quellen - Bestimmung der Massenkonzentration von Stickstoffoxiden (NOx) - Referenzverfahren : Chemilumineszenz

This European Standard was approved by CEN on 30 September 2005.

CEN members are bound to comply with the CEN/CENELEC Internal Regulations which stipulate the conditions for giving this European Standard the status of a national standard without any alteration. Up-to-date lists and bibliographical references concerning such national standards may be obtained on application to the Central Secretariat or to any CEN member.

This European Standard exists in three official versions (English, French, German). A version in any other language made by translation under the responsibility of a CEN member into its own language and notified to the Central Secretariat has the same status as the official versions.

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Foreword

This European Standard (EN 14792:2005) has been prepared by Technical Committee CEN/TC 264 "Air Quality", the secretariat of which is held by DIN.

This European Standard shall be given the status of a national standard, either by publication of an identical text or by endorsement, at the latest by May 2006, and conflicting national standards shall be withdrawn at the latest by May 2006.

This European Standard has been prepared under a mandate given to CEN by the European Commission and the European Free Trade Association, and supports essential requirements of EU Directive(s).

For relationship with EU Directive(s), see informative Annex ZA, which is an integral part of this European Standard.

According to the CEN/CENELEC Internal Regulations, the national standards organizations of the following countries are bound to implement this European Standard: Austria, Belgium, Cyprus, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Netherlands, Norway, Poland, Portugal, Slovakia, Slovenia, Spain, Sweden, Switzerland and United Kingdom.

1 Scope

This European Standard describes the chemiluminescence method, including the sampling and the gas conditioning system, to determine the $NO/NO_2/NO_x$ concentrations in flue gases emitted from ducts and stacks to atmosphere. This European Standard is the Standard Reference Method (SRM) for periodic monitoring and for calibration or control of Automatic Measuring Systems (AMS) permanently installed on a stack, for regulatory or other purposes such as calibration. To be used as the SRM, the user shall demonstrate that the performance characteristics of the method are better than the performance criteria defined in this European Standard and that the overall uncertainty of the method is less than ± 10 % relative at the daily Emission Limit Value (ELV).

NOTE When the chemiluminescence method is the measurement principle used for AMS, reference should be made to EN 14181 and other relevant standards provided by CEN TC 264.

An Alternative Method to this SRM may be used provided that the user can demonstrate equivalence according to the Technical Specification CEN TS 14793, to the satisfaction of his national accreditation body or law.

This SRM has been evaluated during field tests on waste incineration, co-incineration and large combustion installations. It has been validated for sampling periods of 30 min in the range of 0 mg NO_2/m^3 to 1 300 mg NO_2/m^3 for large combustion plants and 0 mg NO_2/m^3 to 400 mg NO_2/m^3 for waste incineration, according to emission limit values (ELVs) laid down in the following Council Directives:

— Council Directive 2001/80/EC on the limitation of emissions of certain pollutants into the air from large combustion plants;

— Council Directive 2000/76/EC on waste incineration plants.

The ELVs for NO_x ($NO + NO_2$) in EU directives are expressed in mg NO_2/m^3, on dry basis, at a reference value for O_2 and at the reference conditions (273 K and 101,3 kPa).

2 Normative references

The following referenced documents are indispensable for the application of this European Standard. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ENV 13005, *Guide to the expression of uncertainty in measurement.*

EN 14790:2003, *Stationary source emissions - Determination of the water vapour in ducts.*

CEN/TS 14793, *Stationary source emissions - Intralaboratory validation procedure for an alternative method compared to a reference method.*

EN ISO 14956, *Air quality - Evaluation of the suitability of a measurement procedure by comparison with a required measurement uncertainty (ISO 14956:2002).*

3 Terms and definitions

For the purposes of this European Standard, the following terms and definitions apply.

NOTE In this European Standard, NO_x is defined as the sum of NO and NO_2. The mass concentration of NO_x is expressed as the equivalent NO_2 concentration in milligrams per cubic metre at normal conditions.

3.1
adjustment (of a measuring system)
operation of bringing a measuring system into a state of performance suitable for its use

[VIM 4.30]

3.2
ambient temperature
temperature of the air around the measuring system

3.3
automatic measuring system (AMS)
measuring system permanently installed on site for continuous monitoring of emissions

NOTE 1 An AM is a method which is traceable to a reference method.

NOTE 2 Apart from the analyser, an AMS includes facilities for taking samples (e.g. probe, sample gas lines, flow meters, regulators, delivery pumps) and for sample conditioning (e.g. dust filter, moisture removal devices, converters, diluters). This definition also includes testing and adjusting devices, that are required for regular functional checks.

[EN 14181]

3.4
calibration
statistical relationship between values of the measurand indicated by the measuring system (AMS) and the corresponding values given by the standard reference method (SRM) used during the same period of time and giving a representative measurement on the same sampling plane

NOTE The result of calibration permits to establish the relationship between the values of the SRM and the AMS (calibration function).

3.5
converter efficiency
percentage of NO_2 present in the sample gas converted to NO by the converter

3.6
drift
difference between two zero (zero drift) or span readings (span drift) at the beginning and at the end of a measuring period

3.7
emission limit value (ELV)
emission limit value according to EU Directives on the basis of 30 min, 1 hour or 1 day

3.8
influence quantity
quantity that is not the measurand but that affects the result of the measurement

[adapted VIM 2.7]

NOTE Examples:

— ambient temperature;

— atmospheric pressure;

— presence of interfering gases in the flue gas matrix;

— pressure of the gas sample.

3.9
interference
negative or positive effect upon the response of the measuring system, due to a component of the sample that is not the measurand

3.10
lack of fit
systematic deviation within the range of application between the measurement result obtained by applying the calibration function to the observed response of the measuring system measuring test gases and the corresponding accepted value of such test gases

NOTE 1 Lack of fit may be a function of the measurement result.

NOTE 2 The expression “lack of fit” is often replaced in everyday language by "linearity" or "deviation from linearity".

3.11
measurand
particular quantity subject to measurement

[VIM 2.6]

3.12
measuring system
complete set of measuring instruments and other equipment assembled to carry out specified measurements

[VIM 4.5]

3.13
performance characteristic
one of the quantities (described by values, tolerances, range…) assigned to equipment in order to define its performance

3.14
repeatability in the laboratory
closeness of the agreement between the results of successive measurements of the same measurand carried out under the same conditions of measurement

NOTE 1 Repeatability conditions include:

— same measurement procedure;

— same laboratory;

— same measuring instrument, used under the same conditions;

— same location;

— repetition over a short period of time.

NOTE 2 Repeatability may be expressed quantitatively in terms of the dispersion characteristics of the results.

In this European Standard the repeatability is expressed as a value with a level of confidence of 95 %.

[VIM 3.6]

3.15
repeatability in the field
closeness of the agreement between the results of simultaneous measurements of the same measurand carried out with two sets of equipment under the same conditions of measurement

NOTE 1 These conditions include:

— same measurement procedure;

— two sets of equipment, the performance of which fulfils the requirements of the reference method, used under the same conditions;

— same location;

— implemented by the same laboratory;

— typically calculated on short periods of time in order to avoid the effect of changes of influence parameters (e.g. 30 min).

NOTE 2 Repeatability may be expressed quantitatively in terms of the dispersion characteristics of the results.

In this European Standard the repeatability under field conditions is expressed as a value with a level of confidence of 95 %.

3.16
reproducibility in the field
closeness of the agreement between the results of simultaneous measurements of the same measurand carried out using several sets of equipment under the same conditions of measurement

NOTE 1 These conditions are called field reproducibility conditions and include:

— same measurement procedure;

— several sets of equipment, the performance of which fulfils the requirements of the reference method, used under the same conditions;

— same location;

— implemented by several laboratories.

NOTE 2 Reproducibility may be expressed quantitatively in terms of the dispersion characteristics of the results.

In this European Standard the reproducibility under field conditions is expressed as a value with a level of confidence of 95 %.

3.17
residence time in the measuring system
time period for the sampled gas to be transported from the inlet of the probe to the inlet of the measurement cell

3.18
response time
time interval between the instant when a stimulus is subjected to a specified abrupt change and the instant when the response reaches and remains within specified limits around its final steady value

NOTE By convention, time taken for the output signal to pass from 0 % to 90 % of the final change

[VIM 5.17]

3.19
sampling location
specific area close to the sampling plane where the measurement devices are set up

3.20
sampling plane
plane normal to the centreline of the duct at the sampling position

[EN 13284-1]

3.21
sampling point
specific position on a sampling line at which a sample is extracted

[EN 13284-1]

3.22
span gas
test gas used to adjust and check a specific point on the response line of the measuring system

NOTE This concentration is often chosen around 80 % of the upper limit of the range or around the ELV.

3.23
standard reference method (SRM)
measurement method recognised by experts and taken as a reference by convention, which gives, or is presumed to give, the accepted reference value of the concentration of the measurand (3.11) to be measured

3.24
uncertainty
parameter associated with the result of a measurement, that characterises the dispersion of the values that could reasonably be attributed to the measurand

3.24.1
standard uncertainty u
uncertainty of the result of a measurement expressed as a standard deviation u

3.24.2
combined uncertainty u_c
standard uncertainty u_c attached to the measurement result calculated by combination of several standard uncertainties according to GUM

3.24.3
expanded uncertainty U
quantity defining a level of confidence about the result of a measurement that may be expected to encompass a specific fraction of the distribution of values that could reasonably be attributed to a measurand

$$U = k \times u$$

NOTE In this European Standard, the expanded uncertainty is calculated with a coverage factor of k = 2, and with a level of confidence of 95 %.

3.24.4
overall uncertainty U_c
expanded combined standard uncertainty attached to the measurement result calculated according to GUM

$$U_c = k \times u_c$$

3.25
uncertainty budget
calculation table combining all the sources of uncertainty according to EN ISO 14956 or ENV 13005 in order to calculate the overall uncertainty of the method at a specified value

4 Principle

4.1 General

This European Standard describes the SRM, based on the chemiluminescence principle for sampling and determining NO_x, NO and NO_2 concentrations in flue gases emitted to atmosphere from ducts and stacks. The specific components and the requirements for the sampling system and the chemiluminescence analyser are described. A number of performance characteristics, together with associated minimum performance criteria are specified for the analyser. These performance characteristics and the overall uncertainty of the method shall meet the performance criteria given in this European Standard. Requirements and recommendations for quality assurance and quality control are given for measurements in the field (see table 1 in 7.2).

4.2 Measuring principle

The principle of chemiluminescence to measure NO_x is based on the following reaction between nitrogen monoxide and ozone:

$$2\,\mathrm{NO} + 2\,\mathrm{O_3} \Rightarrow \mathrm{NO_2} + \mathrm{NO_2}^* + 2\,\mathrm{O_2}$$

$$\mathrm{NO_2}^* \Rightarrow \mathrm{NO_2} + \mathrm{h}\nu$$

Some of the NO_2 created during the reaction of NO and O_3 is in an excited state. When returning to the basic state, these NO_2 molecules can radiate light, the intensity of which depends on the NO content and is influenced by the pressure and presence of other gases.

In a chemiluminescence analyser, gas is sampled through a sampling line and fed at a constant flow rate into the reaction chamber of the analyser, where it is mixed with an excess of ozone for the determination of nitrogen oxide only. The emitted radiation (chemiluminescence) is proportional to the amount of NO present in the sampled gas. The emitted radiation is filtered by means of a selective optical filter and converted into an electric signal by means of a photomultiplier tube.

For the determination of the amount of nitrogen dioxide, the sampled gas is fed through a converter where the nitrogen dioxide is reduced to nitrogen monoxide and analysed in the same way as previously described. The electric signal obtained from the photomultiplier tube is proportional to the sum of concentrations of nitrogen dioxides and nitrogen monoxides. The amount of nitrogen dioxide is calculated from the difference between this concentration and that obtained for nitrogen monoxide only (when the sampled gas has not passed through the converter).

When a dual type analyser is used, both NO and NO_x results are determined continuously. On the contrary, with a single type analyser, the reaction chamber is alternatively fed with the raw gas and with the gas having passed the converter that reduces NO_2 to NO. Therefore, NO and NO_x are determined alternately.

Chemiluminescence analysers are combined with an extractive sampling system and a gas conditioning system. A representative sample of gas is taken from the stack with a sampling probe and conveyed to the analyser through a sampling line and suitable gas conditioning system. The values from the analyser are recorded and/or stored by means of electronic data processing.

Interference due to CO_2 in the sample gas is possible, particularly in the presence of water vapour, due to quenching of the chemiluminescence. The extent of the quenching depends on the CO_2 and H_2O concentrations and the type of analyser used. Any necessary corrections may be made to the analyser output to increase its accuracy for example by reference to correction curves supplied by the manufacturers or by calibrating with gases containing approximately the same concentration of CO_2 as the sample gas.

NOTE 1 Vacuum chemiluminescent NO_x analyser reduces the CO_2 and H_2O quenching error.

NOTE 2 A correction of the NO_x concentration may be necessary if the NH_3 concentration is higher than 20 mg/m^3.

In flue gases from conventional combustion systems the nitrogen oxides consist of more than 95 % NO. The remaining oxide is predominantly NO_2. These two oxides (NO + NO_2) are designated as NO_x.

It should be noted, that in other processes, the ratio of NO to NO_x may be different and other nitrogen oxides may be present.

5 Description of measuring equipment - Sampling and sample gas conditioning systems

5.1 General

A representative volume of flue gas (see 8.2) is extracted from the emission source for a fixed period of time at a controlled flow rate. A filter removes the dust in the sampled volume before the sample is conditioned and passed to the analyser.

Four different sampling and conditioning configurations are available in order to avoid uncontrolled water condensation in the measuring system (see also Annex A). Each of the first three configurations requires the user shall check that the dew point temperature is lower or equal to 4 °C at the outlet of the analyser. The user may correct the results for the remaining water content in order to report results on a dry basis (refer to the table of Annex A in EN 14790:2003).

These configurations are:

— Configuration 1: removal of water vapour by condensation using a cooling system;

— Configuration 2: removal of water vapour through elimination within a permeation drier;

— Configuration 3: dilution with dry, clean ambient air or nitrogen of the gas to be characterised;

— Configuration 4: maintaining the temperature of the sampling line up to the heated analyser.

It is important that all parts of the sampling equipment upstream of the analyser are made of materials that do not react with or absorb NO_x. Except for the cooling system of configuration 1, the temperature of the components likely to be in contact with the gas, shall be maintained at a sufficiently high temperature to avoid any condensation.

Conditions and layout of the sampling equipment contribute to the overall uncertainty of the measurement. In order to minimise this contribution to the overall measurement uncertainty, performance criteria for the sampling equipment and sampling conditions are specified in sections 5.2 and 7.2.

In order to minimise losses of NO_x in the sampling system, the use of configuration 1 shall be avoided when the measured ratio NO_2/NO_x represents more than 10 %.

5.2 Sampling line components

5.2.1 Sampling line

In order to access the representative measurement point(s) of the sampling plane, probes of different lengths and inner diameters may be used. The design and configuration of the sampling line used shall ensure the residence time of the sample gas is minimised in order to reduce the response time of the measuring system.

The procedure of clause 8.2 shall be used when the operator suspects that the flue gas is inhomogeneous.

NOTE 1 The probe may be marked before sampling in order to demonstrate that the representative measurement point(s) in the measurement plane has (have) been reached.

NOTE 2 A seal-able connection may be installed on the probe in order to introduce test gases for adjustment.

The line shall be made of suitable corrosion resistant material (e.g. stainless steel, borosilicate glass, ceramic ; PTFE is only suitable for flue gas temperature lower than 200 °C). At temperatures greater than 250 °C, stainless steel and certain other materials can alter the ratio of NO_2/NO_x. In this case, ceramic, glass, quartz or titanium should be used. The use of any materials made from copper or copper based alloys are not permitted.

5.2.2 Filter

The particle filter shall be made of an inert material (e.g. ceramic or sinter metal filter with an appropriate pore size). It shall be heated above the water or acid dew point. The particle filter shall be changed or cleaned periodically depending on the dust loading at the sampling site.

NOTE Overloading of the particle filter may cause loss of nitrogen dioxide by sorption onto the particulate matter and may also increase the pressure drop in the sampling line.

5.2.3 Sample cooler (configuration 1)

The design of the sample gas cooler shall be such that absorption of NO_2 in the condensates is minimised. Because overpressure in the cooling system increases losses of NO_2 in the condensates, the pump shall be situated between the cooling system and the analyser. In the case that the concentration of NO_2 in the sample gas becomes too high, the use of a gas cooler can produce errors on the NO_2 measurement. This can occur because of the solubility of NO_2 in the condensed water and shall also depend on the content of water vapour in the flue gas. A maximum dew-point temperature of 4 °C shall not be exceeded at the outlet of the sample cooler.

In order to minimise losses of NO_x in the sampling system, the use of configuration 1 shall be avoided when the measured ratio NO_2/NO_x represents more than 10 %.

NOTE The concentrations, provided by this sampling configuration, are considered to be given on dry basis. However, the user may correct the results for the remaining water (refer to the table of Annex A in EN 14790:2003).

5.2.4 Permeation drier (configuration 2)

The permeation drier is used before the gas enters the analyser in order to separate water vapour from the flue gas. The dew point at the outlet of the system shall be sufficiently below the ambient temperature. A dew-point temperature of 4 °C shall not be exceeded at the outlet of the permeation drier.

NOTE The concentrations, provided by this sampling configuration, are considered to be given on dry basis. However, the user may correct the results for the remaining water (refer to the table of Annex A in EN 14790:2003).

5.2.5 Dilution system (configuration 3)

The dilution technique is an alternative to hot gas monitoring or sample gas drying. The flue gas is diluted with dry, clean, ambient air or nitrogen. The dilution gas shall be dry and free from nitrogen oxides. The dilution ratio shall be chosen according to the objectives of the measurement and shall be compatible with the range of the analytical unit. It shall remain constant through the period of the test. The water dew point shall be reduced so to avoid the risks of condensation. The dew point temperature at the outlet of the analyser shall be determined in order to correct the results and give them on a dry basis (refer to the table of Annex A of EN 14790:2003) if the dew-point temperature is higher than 4 °C.

NOTE Analysers that are used in combination with dilution probes, work with measuring ranges, which are typical for ambient air analysers (0 mg/m^3 –1 mg/m^3 - 5 mglm^3 - 10 mg/m^3 - 25 mg/m^3).

5.2.6 Heated line and heated analyser (configuration 4)

To avoid condensation the user shall maintain the temperature of the sampling line up to the measuring cell. The analyser itself is heated.

The concentrations are given on wet basis and shall be corrected so that they are expressed on dry basis. The correction shall be made from the water vapour concentration measured in the flue gases and the uncertainty attached to this correction shall be added to the uncertainty budget (see clause 7).

5.2.7 Sample pump

When a pump is not an integral part of the chemiluminescence analyser, an external pump is necessary to draw the sampled air through the apparatus. It shall be capable of operating to the specified flow requirements of the manufacturer of the analyser and pressure conditions required for the reaction chamber. The pump shall be resistant to corrosion and shall be of materials that do not react with or absorb NO_x. It shall be consistent with the requirements of the analyser to which it is connected.

NOTE The quantity of sample gas required can vary between 15 l/h and 500 l/h, depending upon the analyser and the expected response time.

5.2.8 Secondary filter

The secondary filter is used to separate fine dust, with a pore size of 1 µm to 2 µm. For example it may be made of glass-fibre, sintered ceramic, stainless steel or PTFE-fibre.

NOTE No additional secondary filter is necessary when they are part of the analyser itself.

5.2.9 Flow controller and flow meter

This apparatus sets the required flow. A corrosion resistant material shall be used. The sample flow rate into the instrument shall be maintained according to the analyser manufacturer's requirements. A controlled pressure drop across restrictors is usually employed to maintain flow rate control into the chemiluminescence analyser.

NOTE No additional flow controller or flow meter is necessary when they are part of the analyser itself.

6 Analyser equipment

6.1 General

Generally two configurations of analysers are available:

— dual type with two reaction chambers;

— single type with one reaction chamber.

In the dual-chamber types the sample flow is divided into two streams, one passing directly to one of the reaction chambers for measurement of the nitrogen monoxide content. The other stream is fed through the converter for conversion of the nitrogen dioxide to nitrogen monoxide and then to the other reaction chamber for measurement of the total content of nitrogen dioxide and nitrogen monoxide.

In the single-chamber type analyser the sample alternately bypasses or passes through the converter. This type of analyser sequentially measures the nitrogen monoxide of the sampled gas followed by the sum of nitrogen dioxide and nitrogen monoxide. If NO_2 and NO shall be measured, the cycle rate of switching between NO_x (through the converter) and NO (bypassing the converter) shall be faster than the rate of change of the NO_x in the sample stream (otherwise the subtraction can produce negative NO_2 readings).

A chemiluminescence apparatus consists of the principal components described in paragraphs 6.2 to 6.7.

6.2 Converter

An NO_2/NO converter shall be included in the measuring system. The converter converts nitrogen dioxide in the gas to nitrogen monoxide. It shall be capable of converting at least 95 % of the nitrogen dioxide to nitrogen monoxide. The converter efficiency shall be determined by one of the procedures described in Annex B. The converter shall consist of a heated furnace maintained at a constant temperature and is made of material such as stainless steel, molybdenum, tungsten, spectroscopicaly pure carbon or quartz.

NOTE 1 The converter can be bypassed with a changeover valve arrangement. If the sample gas flows through the converter, the total quantity (NO + NO_2) is determined; when the converter is by-passed, the NO content is determined. The amount of NO_2 can be calculated as the difference between NO_x and NO.

NOTE 2 If ammonia is present in the sample gas, interferences can occur depending on the operating temperature and the material of the converter.

NOTE 3 When using a stainless steel converter in a sample that is a reducing atmosphere, the NO in the sample can be disassociated to N_2 and O_2. This leads to lower readings. Manufacturers of analysers that use stainless steel normally have an air bleed from the ozonator supply to the sample stream in order to avoid this effect.

Examples of different types of converters are given in Annex C.

6.3 Ozone generator

An ozone generator is required. Ozone is generated from oxygen by either ultra-violet radiation or by a high-voltage silent-electric discharge. If oxygen in ambient air is used for ozone generation by a high-voltage silent-electric-discharge, it is essential that the air is thoroughly dried and filtered before entering the generator. If the ozone is generated using oxygen of a recognised analytical grade from a compressed gas cylinder, this oxygen can be fed directly into the generator. The concentration of ozone produced shall be sufficiently high to maintain the required linearity of the analyser. If ozone concentration is too low, this can result in a non-linear response to the concentration of NO_2 and NO. It may be useful to check that the quantity of O_3 is sufficient by injecting a calibration gas of NO, which concentration is close to the upper limit of the measurement range.

WARNING — In order to minimise the risk of explosions, the necessary precautions shall be taken and safety regulations shall be applied when oxygen is used instead of air.

The flow rate of air or oxygen shall fulfil specifications given by the manufacturer of the analyser.

6.4 Reaction chamber

The reaction chamber shall be constructed of an inert material. Its dimensions determine the characteristics of the chemiluminescence reaction (residence time, speed of reaction). The reaction chamber shall be heated at a constant temperature above the dew point of the sampled gases at the pressure conditions in this chamber.

The chemiluminescence reaction is carried out at reduced pressure to minimise quenching effects (interferences) and to increase sensitivity.

6.5 Optical filter

This filter shall remove radiation at wavelengths shorter than 600 nm, to minimise any interference produced by the chemiluminescence reaction with unsaturated hydrocarbons that radiate at these wavelengths.

6.6 Photomultiplier tube

The output of the analyser is affected by the characteristics of the photomultiplier tube(s). In order to reduce background noise and the effect of temperature changes, the tube is generally housed in a thermostatically controlled refrigerated device (a Peltier cooler is commonly used.)

6.7 Ozone removal

The ozone shall be removed from the gas on leaving the reaction chamber by passage through an ozone removal device. This prevents contamination of the immediate surrounding air and protects the sampling pump.

7 Determination of the characteristics of the SRM: analyser, sampling and conditioning line

7.1 General

When this European Standard is used as the SRM, the user shall demonstrate that:

— performance characteristics are better than the minimum performance criteria given in Table 1; and

— overall uncertainty calculated by combining values of standard uncertainties associated to the performance characteristics given in Table 1 is less than ± 10 % at the daily ELV expressed on dry basis and before correction to the O_2 reference concentration.

The values of the selected performance characteristics shall be evaluated by means of a laboratory test and a field test. An experienced laboratory recognised by the competent authority shall perform the laboratory tests according to the procedures described in the relevant CEN/TC264 standards or technical specifications. The user shall perform the field tests.

NOTE The performance characteristics and performance criteria given in this European Standard are not necessarily corresponding to those of an AMS. For AMS performance characteristics and performance criteria the user should refer to the relevant standards provided by CEN/TC 264.

7.2 Relevant performance characteristics of the SRM and performance criteria

The uncertainty of the measured values given by the method is not only influenced by the performance characteristics of the analyser itself but also by:

- sampling system including conditioning;
- site specific conditions;
- calibration gases used.

Table 1 gives an overview of the relevant performance characteristics and performance criteria, which shall be determined during laboratory and field tests according to the relevant CEN procedures, and indicates values included in the calculation of the overall uncertainty.

Table 1 — Minimum performance characteristics of the SRM

Performance characteristics	Lab. test for the analyser	Field test	Performance criterion	Performance characteristics to be included in calculation of overall uncertainty
Response time	X	X	≤ 200 s	
Detection limit	X		≤ ± 2,0 % of the range	
Lack of fit	X		≤ ± 2,0 % of the range	X
Zero drift	X		≤ ± 2,0 % of the range/24 h	X
Span drift	X		≤ ± 2,0 % of the range/24 h	X
Sensitivity to atmospheric pressure	X		≤ ± 3,0 % of the range/ 2 kPa	X
Sensitivity to sample volume flow or sample pressure	X		[a]	X
Sensitivity to ambient temperature	X		≤ ± 3,0 %of the range/10 K	X
Sensitivity to electric voltage	X		≤ ± 2,0 % of the range/10 V	X
Interferents [b]	X		Total ≤ ± 4,0 % of the range	X
Converter efficiency	X		≥ 95,0 %	X
Losses and leakage in the sampling line and conditioning system		X	≤ ± 2,0 % of the measured value	X
Standard deviation of repeatability in laboratory at zero	X		≤ ± 1,0 % of the range	X [c]
Standard deviation of repeatability in laboratory at span level	X		≤ ± 2,0 % of the range	X [c]

[a] The tested volume flow range or pressure is defined in the manufacturer's recommendations.

[b] Interferents that shall be tested are at least those given in table 2.

The value of max algebric (sum of contributions to uncertainty producing positive effects on the result, sum of contributions to uncertainty producing negative effects) shall be compared with the performance criterion.

[c] Only one of both values shall be included in the calculation: the first possibility is to choose the repeatability standard deviation obtained from laboratory tests corresponding to the closest concentration to the actual concentration in stack, or the higher (relative) standard deviation of repeatability independently of the concentration measured in stack.

7.3 Establishment of the uncertainty budget

An uncertainty budget shall be established to determine if the analyser and its associated sampling system fulfil the requirements for a maximum allowable overall uncertainty.

The overall uncertainty for this method used as a reference shall be lower than 10 % relative at the daily ELV. This overall uncertainty is calculated on dry basis and before correction to the O_2 reference concentration.

The principle of calculation of the overall uncertainty is based on the law on propagation of uncertainty laid down in ENV 13005:

— determine the standard uncertainties for each value included in the calculation of the uncertainty budget by means of laboratory and field tests, and according to ENV 13005;

— calculate the uncertainty budget by combining all the standard uncertainties according to ENV 13005, including the uncertainty of the calibration gas and taking variations range of influence quantities and interferents of the specific site conditions into account. If these conditions are unknown, default values defined in Table 2 shall be applied. For configuration 4, additional sources of uncertainty coming from the dilution system shall be added to the uncertainty budget and the uncertainty attached to the correction of water vapour content shall be added to the uncertainty budget;

— values of standard uncertainty that are less than 5 % of the maximum standard uncertainty can be neglected;

— calculate the overall uncertainty at the daily ELV, reported as a dry gas value.

Table 2 — Default variations ranges of influence quantities and values of interferents to be applied for the uncertainty budget

Influence quantity	Variations range on site
Atmospheric pressure	± 2 kPa
Sample volume flow or pressure variation	In accordance with the manufacturer's recommendations
Ambient temperature	± 15 °C
Electric voltage at span level	230 V ± 20V
CO_2	15 % volume
NH_3	20 mg/m^3
H_2O[a]	20 % volume
[a] Only for sampling without any removal of water vapour	

An example of an uncertainty budget is given in Annex D.

8 Field operation

8.1 Sampling location

The sampling location chosen for the measurement devices and samplings shall be of sufficient size and construction, that a representative emission measurement can be made that is suitable for the measurement task and technically perfect if possible. In addition, the sampling location shall be chosen with regard to safety of the personnel, accessibility and availability of electrical power.

8.2 Sampling point(s)

The gas concentrations measured shall be representative of the average condition of the waste gas. In most cases, a single sampling point situated in the middle of the duct shall be selected. For larger ducts, this point may be situated close to the sampling port provided that there is no disturbance of the flow or concentration due to the influence of the sampling port.

However, when non-homogeneity of the flue gas is suspected, it shall be demonstrated if the flue gas is homogeneous or not. The homogeneity can be demonstrated with a continuous measurement of O_2 or CO_2 using a fixed and a moving probe. Sampling in non-homogeneous flue gases requires following the relevant standards or technical specifications proposed by CEN TC 264.

8.3 Choice of the measuring system

The following flue gas characteristics shall be determined, before the field campaign, so that an appropriate analyser and sampling line configuration (including conditioning unit) can be adopted:

— temperature of exhaust gases;

— flue gas moisture content;

— dust loading;

— expected concentration range of NO_x and ELVs;

— NO_2/NO_x ratio;

— expected concentration of potentially interfering substances, including at least the three components listed in Table 2.

The full scale of the selected analyser shall be 150 % to 300 % of the half hourly ELV and shall not be less than peak emission.

The sampling line should be as short as possible to avoid long response times. If necessary a bypass pump together with a heated filter appropriate to the dust loading shall be used.

The laboratory shall verify that the analyser is functioning in accordance with the required performance criteria fixed in this European Standard, and that the sampling line and conditioning unit are in good operational conditions before conducting field measurements.

8.4 Setting of the SRM on site

8.4.1 General

The complete measuring system, including the conditioning unit, the sampling line and the analyser, shall be connected according to the manufacturer's instructions and the nozzle of the probe placed at the representative point(s) in the duct (see 8.2).

The conditioning unit, sampling probe, filter, connection tube and analyser shall be stabilised at the required temperature. At the same time, a constant low pressure shall be achieved in the measuring chamber of the analyser.

After pre-heating, the flow passing through the sampling system and the analyser shall be adjusted to the chosen flow rate to be used during measurement. This flow shall be maintained at a constant level (± 10%).

When both the instrument and sampling system have been set-up, the user shall check that the instrument and sampling system are functioning correctly. The results of these checks shall fulfil the requirements and limitations as set out by the manufacturer of the instrument, as well as the requirements (such as materials used, and so on) given by this European Standard. Compliance with the requirements of this European Standard shall be documented.

Any data recording, data processing and telemetry system used in conjunction with the measuring system shall be checked for proper functioning. If any components are changed, then these checks shall be repeated. All checks shall be documented. The time resolution of the data recording system used to calculate the mean values shall be below or equal to 1min.

8.4.2 Preliminary zero and span check, and adjustments

8.4.2.1 Test gases

8.4.2.1.1 Oxygen

Oxygen or air, both purified, to provide the ozone generator according to the manufacturers instructions.

8.4.2.1.2 Zero gas

The zero gas shall be a gas containing no significant amount of nitrogen oxides (for example, purified nitrogen or air).

8.4.2.1.3 Span gases

The span gases used to adjust the instrument shall have a certified concentration of NO or NO/NO_2. The uncertainty on the analytical certificate of the span gas shall be less than or equal to $\pm$ 2 % for NO or NO/NO_2.

When the analyser is used for regulatory purposes, the span gas shall have a known concentration of approximately the half-hourly ELV or 50 % to 90 % of the selected range of the analyser.

8.4.2.2 Adjustment of the analyser

At the beginning of the measuring period, zero and span gases are supplied to the analyser directly, without passing through the sampling system. Adjustments are made until the correct zero and span gas values are given by the data sampling system:

— adjust the zero value;

— adjust the span;

— finally, re-check zero to see if there are no significant changes (zero deviation lower than 2 times the repeatability at zero). If there is any problem, repeat the procedure.

8.4.2.3 Check of the sampling system

Before starting the measurement, one of the following procedures shall be applied to determine if there is leakage in the sampling line:

— Zero and span gas are supplied to the analyser through the sampling system, as close as possible to the nozzle (in front of the filter if possible). Any differences between the readings obtained during the adjustment of the analyser and during the check shall be lower than 2 %.

NOTE Differences may be due to leakage or impurities in the sampling system leading to memory effects due to adsorption or desorption to or from the surfaces.

— Check the sampling line for leakage according to the following procedure or any other relevant procedure:

Assemble the complete sampling system. Close the nozzle and switch on the pump. After reaching minimum pressure, measure the flow rate with an appropriate measuring device. The leak flow rate shall not exceed 2 % of the expected sample gas flow rate used during measurement.

8.4.3 Zero and span checks after measurement

At the end of the measuring period and at least once a day, zero and span checks shall be performed at the inlet of the sampling system by supplying test gases. The results of these checks shall be documented. In case of deviation between checks after measurement and preliminary adjustments, values of deviation shall be indicated in the report.

If the span or zero drifts are bigger than 2 % of the span value, it is necessary to correct both for zero and span drifts (see in Annex E for a procedure of correction of data for drift effect).

The drift of zero and span shall be lower than 5 % of the span value; otherwise, the results shall be rejected.

9 Ongoing quality control

9.1 General

Quality control is critically important in order to ensure that the uncertainty of the measured values for NO and NO_2 are kept within the stated limits during extended automatic monitoring periods in the field. This means that maintenance, as well as zero and span adjustment procedures shall be followed, as they are essential for obtaining accurate and traceable quality data.

9.2 Frequency of checks

Table 4 shows the minimum required frequency of checks. The laboratory shall implement the relevant European Standards for determination of performance characteristics and procedures described in Annex B.

Table 3 — Frequency of checks as reference method

Checks	Frequency	Action criteria
Converter efficiency determination (6.2)	At least every year[a]	Efficiency < 95,0 %
Cleaning or changing of particulate filters [b] at the sampling inlet and at the monitor inlet	Every campaign if needed [b]	
Checking or changing of ozone removal device (6.7), and if applicable: drying material and other consumables	At least every year [c]	As required by the manufacturer
Regular maintenance of several parts of the monitor	As required by manufacturer	As required
Linearity check	At least every year and after repair of the instrument	As required and when linearity > ± 2,0 % of the range

[a] Depending on the converting material and on NO_2 concentration.

b The particle filter shall be changed periodically depending on the dust loading at the sampling site. During this filter change the filter housing shall be cleaned. Overloading of the particle filter may cause losses of nitrogen dioxide by sorption on the particulate matter and may increase the pressure drop in the sampling line.

[c] Strongly dependent on site-specific conditions.

10 Expression of results

The readings from the analyser are converted to concentrations using the appropriate calibration function and the results are expressed in milligrams per cubic metre, or parts per million by volume, on dry basis at 273 K and 101,3 kPa.

For nitrogen monoxide (NO) the conversion factors at 273 K and 101,3 kPa are:

1 mg NO/m^3 = 0,74 ppm (V/V) NO

1 ppm (V/V) NO = 1,34 mg NO/m^3 or 2,05 mg NO_2/m^3

For nitrogen dioxide (NO_2) the conversion factors at 273 K and 101,3 kPa are:

1 mg NO_2/m^3 = 0,487 ppm (V/V) NO_2

1 ppm (V/V) NO_2 = 2,05 mg NO_2/m^3

When results are given on wet basis, the concentration on dry basis are calculated according to the following equation:

$$C_d = \frac{100}{100 - H_2O} \times C_w \quad (1)$$

where

Cd is the concentration of the measurand on dry basis

Cw is the measured concentration of the measurand on wet basis

H_2O is the water vapour content measured in gas

Concentrations are generally expressed at a reference concentration of O_2 defined in the European Directives:

— 11% for incineration of waste in waste incinerators;

— 10 % for co-incineration of waste in cement kilns;

— 3% for combustion of gas or liquid fuels;

— 6% for combustion of solid fuels;

— 15% for gas turbines.

The corrected concentration of measurand C_{corr} is calculated using the followed equation:

$$C_{corr} = \frac{21 - O_{2,ref}}{21 - O_{2,mes}} \times C_{actual} \quad (2)$$

where

C_{corr} is the corrected concentration of measurand

C_{actual} is the measured concentration of measurand at actual O_2 concentration

$O_{2,mes}$ is the measured mean dry oxygen content during the sampling time

$O_{2,ref}$ is the oxygen reference concentration

11 Evaluation of the method in the field

The method has been evaluated during six field tests, on waste incineration installations, co-incineration installations and large combustion plants. Each test was performed by at least four different measuring teams originating from ten European Union member states.

The characteristics of installations, the conditions during field tests and the values of repeatability and reproducibility in the field are given in Annex F.

12 Equivalence with an alternative method

In order to show that an alternative method is equivalent to the reference method, follow the procedures described in the CEN TS 14793.

The maximum allowable standard deviation of repeatability and reproducibility expressed in mg/m^3 for this reference method is:

$$S_{r\,\mathrm{limit}}(C) = S_R(C) = 0{,}0153\,C + 2{,}1 \quad (3)$$

where

C is the concentration in milligrams per cubic meter.

13 Test report

The test report shall include the following information:

a) Reference to this European standard;

b) Description of the purpose of tests;

c) Information about the personnel involved in the measurement;

d) Principle of gas sampling and conditioning;

e) Information about the analyser and description of the sampling and conditioning line;

f) Identification of the analyser used, and analyser’s performance characteristics as listed in table 1;

g) Operating range;

h) Details of the quality and the concentration of the calibration gases used;

i) Details on the adjustment performed before and after actual sampling (at the inlet of the sampling line and at the inlet of the analyser);

j) Description of plant and process;

k) Identification of the sampling plane;

l) Description of the location of the sampling point(s) in the sampling plane;

m) Description of the operating conditions of the plant process;

n) Changes in the plant operations during sampling, for example burner changes;

o) Measurement results with sampling date, time and duration;

p) Information on flue gas characteristics (temperature, velocity, moisture, pressure);

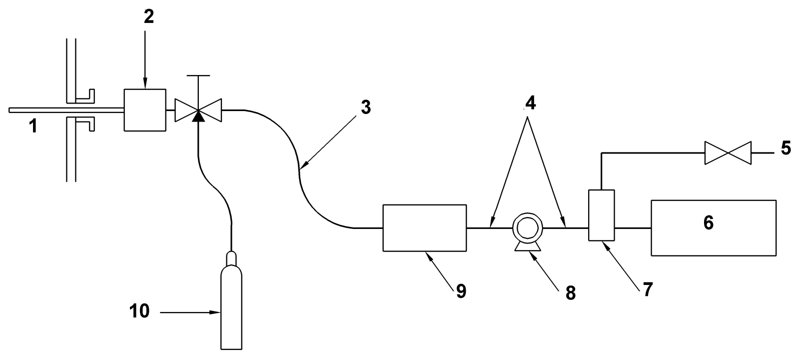
q) Time averaging on relevant periods;

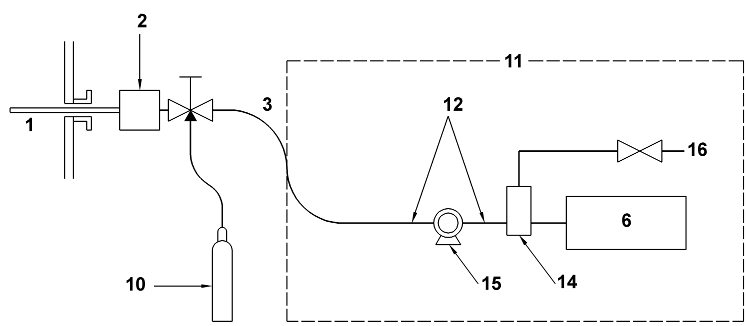
r) Uncertainty budget established according to EN/ISO 14956 or ENV 13005;

s) Any deviations from this European Standard.

Annex A
(informative)

Four different sampling and conditioning configurations





Key

1 Stack
2 Heated filter
3 Heated sample line
4 Sample gas transport line (PTFE)
5 Sample by-pass vent
6 Gas analyser
7 Sample gas manifold
8 Sample pump
9 Conditioning system :
configuration 1: condenser with a cooling system
configuration 2: permeation drier
configuration 3: dilution stage
10 Calibration gas(es)
11 Heated and temperature controlled zone (configuration 4)
12 Gas transport line (PTFE)
13 by-pass vent
14 gas manifold
15 pump

Figure A.1

Annex B
(normative)

Determination of converter efficiency

B.1 General

The converter efficiency is tested by means of measurements with known concentrations of NO_2. The NO_2 is delivered using a cylinder which contains a carrier gas such as nitrogen, and a known concentration of NO_2.

B.2 First method : cylinder gases for calibration

For analysers with two reaction chambers, adjust the analyser on the NO and NO_x channels with a NO concentration approximately equal to the ELV, ensuring that both channels give the same value. Then record the values.

Inject the analyser with a known concentration of NO_2.

Record the NO_x and NO concentration. Wait at least six response times between the readings of the NO_x and NO concentrations. A measurement shall be taken as the average of a time period of at least three response times.

Calculate the converter efficiency by the following equation:

$$Conv.Eff.(\%) = \frac{(NO_x)_u - (NO)_u}{(NO_2)_i} \times 100\ \% \tag{B.1}$$

where

Conv.Eff. (%)	is the converter efficiency in percentage
$(NO_x)_u$	is the NO_x reading with NO_2 containing gas
$(NO)_u$	is the NO reading with NO_2 containing gas
$(NO_2)_i$	is the NO_2 concentration of the NO_2 containing gas applied to the analyser

Perform the determination of the converter efficiency three times and report the average of the three determinations.

B.3 Second method : gaseous phase titration

The following equipment is required:

— source of nitrogen monoxide such as a compressed gas cylinder containing nitrogen monoxide in nitrogen at a concentration of the order of 80 % of the certification range. The actual concentration does not need to be known provided that it remains constant throughout the test;

— source of oxygen, such as a compressed gas cylinder containing air or oxygen;

— ozone generator, capable of providing varying amounts of ozone from oxygen.

NOTE 1 When using a corona-discharge ozone generator, pure oxygen should be used instead of air. Otherwise the corona-discharge ozone generator may oxidise the N_2 in the air to NO and result in a positive bias to the converter efficiency calculations.

Ensure that the total flow rate of nitrogen monoxide and air (or oxygen) is greater than the flow rate of gas through the analyser.

In all of the following steps, determine the responses of the analyser to both the nitrogen monoxide and total NO_x.

NOTE 2 This procedure evaluates the concentration of nitrogen dioxide being produced, which should be in the range 10 % to 90 % of the NO_x.

— Turn off the ozone generator. Then record the concentration of total NO_x (R_1) and the concentration of nitrogen monoxide (P_1);

— Turn on the ozone generator. Record the concentration of total NO_x (R_2) and the concentration of nitrogen monoxide (P_2).

NOTE 3 Ozone is formed which reacts with the nitrogen monoxide to produce nitrogen dioxide before the gases enter the analyser.

— Vary the output of the ozone generator, and record the displayed concentrations of total NO_x (R_2, R_3, R_4,…R_n) and nitrogen monoxide (P_2, P_3, P_4, …, P_n).

NOTE 4 The ratios R_2/R_1, R_3/R_1, R_4/R_1, … Rn/R1 should be as close to unity as possible. Therefore the concentration of total NO_x should be constant in each case and independent of the ratio of the concentrations of nitrogen dioxide to nitrogen monoxide.

— Calculate the converter efficiency, expressed as a percentage, using equation B.2. R_n and P_n are the recorded concentrations of total NO_x and nitrogen monoxide for each setting of the ozone generator.

$$C_E = \frac{(R_n - P_n) - (R_1 - P_1)}{P_1 - P_n} x\, 100 \qquad \text{(B.2)}$$

where

C_E is the converter efficiency

$R_{1,\,2,\,\dots n}$ is the concentration of total NO_x

$P_{1,\,2,\,\dots\, n}$ is the concentration of NO

Annex C
(informative)

Examples of different types of converters

C.1 Quartz converter

A silica coil heated and maintained at around 850 °C is used. The conversion efficiency is normally higher than 96 %.

Advantages: almost no maintenance.

Drawbacks: an unwieldy device, which needs to be thermally isolated to avoid the radiance effects.

Consumption of electricity.

Connections between metal and glass to be verified periodically.

C.2 Low temperature converter (molybdenum)

A cartridge in stainless steel with molybdenum cuttings, from which the grease has been removed, heated and adjusted at around 380 °C. The converter efficiency should be above 97 %. The cuttings are often replaced every 6 months.

C.3 Stainless steel converter

A 316 L grade stainless steel coil is used. It is heated and maintained at around 650 °C. The conversion efficiency is normally higher than 97 %.

NOTE This type of converter can be combined with a molybdenum converter to measure NH_3.

Annex D
(informative)

Example of assessment of compliance of chemiluminescence method for NO_x with requirements on emission measurements

D.1 General

This informative Annex gives an example of calculation of the overall uncertainty established with configuration 1.

D.2 Process of uncertainty estimation

The procedure for calculating measurement uncertainty as follows is based on the law on propagation of uncertainty laid down in EN ISO 14956 or ENV 13005. The calculation procedure presents different steps:

D.2.1 Determination of the model equation

Define the measurand and all the parameters that influence the result of the measurement. These parameters, called “input quantities” shall be clearly defined.

Identify all sources of uncertainty contributing to any of the input quantities or to the measurand directly.

Then the model equation, that is to say the relationship between the measurand and the influence quantities, shall be established, if possible in mathematical equation form.

D.2.2 Quantification of uncertainty components

Each uncertainty source is estimated to obtain its contribution to the overall uncertainty.

Use available performance characteristics of the measurement system, data from the dispersion of repeated measurements, data provided in calibration certificates …

Convert all uncertainty components (e.g. performance characteristics) to standard uncertainties of input and influence quantities.

D.2.3 Calculation of the combined uncertainty

Then the combined uncertainty u_c is calculated by combining standard uncertainties, by applying the “law of propagation of uncertainty”.

In general, the uncertainty associated to a concentration is expressed in overall uncertainty form. The overall uncertainty U_c corresponds to the expanded combined uncertainty, obtained by multiplying by a coverage factor “k”: $U_c = k \times u_c$. The value of the coverage factor *k* is chosen on the basis of the level of confidence required. In most cases, *k* is taken equal to 2, for a level of confidence of approximately 95 %.

The following Tables give one example of:

— Specific conditions of the site (Table D.1), that is to say the values and the variation range of the influence parameters.

— Performance characteristics of the method (Table D.2) related to the parameters which can have an influence on the response of the analyser: the performance characteristics of the analyser and the effect of influence parameters (e.g. chemical interferents, and environmental conditions such as ambient temperature, voltage and pressure).

D.3 Specific conditions in the field

Table D.1 — Example of site specific conditions

Specific conditions	Value / range
Range of analyser	0 ppm to 200 ppm
Studied concentration of NO_x (limit value of NO_x for the site)	200 mg/m^3 eq NO_2 at $O_{2,ref}$ (corresponding to: ≈ 97 ppm at $O_{2,ref}$)
Conditions on the field Ratio NO/NO_x Sample volume flow Temperature during adjustment Fluctuations of ambient temperature during measurement Voltage variation Atmospheric pressure during adjustment Atmospheric pressure variation	 94 % 60 l/h ± 5 l/h 285 K 283 K to 308 K 230V ± 5 % 99 kPa 99 kPa to 100 kPa
O_2 reference concentration: $O_{2,ref}$	11 % volume
NH_3 concentration variations in the field CO_2 concentration variations in the field	0 mg/m^3 to 20 mg/m^3 8 % to 15 %
Test gas NO in N_2, without interferent	180 ppm ± 2 %

D.4 Performance characteristics of the method

Concentration range of analyser during type approval tests: 200 ppm

Number of measurement cell of the analyser: 1

Table D.2 — Example of performance characteristics

Performance characteristics for chemiluminescence method	Performance criteria	Results laboratory and field tests
Response time	≤ 200 s	120 s
Detection limit	≤ ± 2,0 % of the range	± 1,3 % of the range
Lack of fit	≤ ± 2,0 % of the range	± 0,7 % of the range
Zero drift	≤ ± 2,0 % of the range / 24 h	± 0,01 % of range/24 h
Span drift	≤ ± 2,0 % of the range / 24 h	± 1 % of the range/24 h
Sensitivity to the sample volume flow	≤ 1,0 % of the range	1 % of the range/10 l/h
Sensitivity to atmospheric pressure	≤ 3,0% of the range/2 kPa	1,6 % of the measured value/2 kPa
Sensitivity to ambient temperature	≤ 3,0 % of the range /10 K	1 % of range/10 K
Sensitivity to electric voltage at span level	≤ 2,0 % of the range/10 V	0,12 % of range/10 V
Interferents NH_3 (20 mg/m^3) CO_2 (15%)	Total ≤ ± 4,0 % of the range	 0,75 ppm -2,6 ppm
Converter efficiency Drift of converter efficiency between 2 calibrations Standard deviation of repeatability of converter efficiency	≥ 95,0 %	98 % 3 % (absolute) 1,0 % (absolute)
Standard deviation of repeatability in laboratory at zero	≤ 1,0 % of the range	0,65 % of range
Standard deviation of repeatability in laboratory at span level	≤ 2,0 % of the range	0,8 % of range

Calculation of standard uncertainty of concentration values given by the analyser.

Model equations in this clause, as well as calculations of partial uncertainties are related to the values measured by the analyser and expressed in ppm.

Uncertainty associated to conversions in mg/m^3 is handled later in clause D.4.

D.4.1 NO measurement

D.4.1.1 Model equation and application of rule of uncertainty propagation

The concentration in NO is equal to the concentration given by the analyser plus corrections due to deviations associated to influence quantities and to the performance characteristics of the analyser.

$$C_{NO,ppm} = C_{NO,read} + Corr_{fit} + Corr_{0,dr} + Corr_{s,dr} + Corr_{rep} + Corr_{adj} + \sum_{j=1}^{p} Corr_{\inf} + Corr_{\text{int}} \quad \text{(D.1)}$$

where

$C_{NO,ppm}$	is the concentration of NO in ppm
$C_{NO,read}$	is the concentration of NO given by the analyser
$Corr_{fit}$	is the correction of lack of fit
$Corr_{0,dr}$, $Corr_{s,dr}$	are the corrections of zero drift and span drift
$Corr_{rep}$	is the correction of repeatability of the measurement
$Corr_{adj}$	is the correction of adjustment
$Corr_{\inf}$	is the correction of influence quantities (ambient temperature, atmospheric pressure, sample volume flow, voltage)
$Corr_{\text{int}}$	is the correction of interferents

The corrections are specific to each analyser. Sometimes the value of correction is nil. For example, influence parameters, can increase or decrease during the measurement period and are generally not monitored; therefore, it can be considered that the best correction to be applied is zero.

Whether the corrections are equal to zero or not, the uncertainties associated with these corrections shall be taken into account in the calculation of the global uncertainty. The combined standard uncertainty associated to NO measurement is given by:

$$u^2(C_{NO,ppm}) = \sum_{i=1}^{N} \left[\left(\frac{\partial C_{NO,ppm}}{\partial Xi} \right)^2 \times u^2(Xi) \right] \quad \text{(D.2)}$$

where

$u(C_{NO,ppm})$	is the combined standard uncertainty associated to NO measurement
Xi, i =1 to N	are the influence quantities
$\dfrac{\partial C_{NO,ppm}}{\partial Xi}$	is the sensitivity coefficient of $C_{NO,ppm}$, to Xi
$u(Xi)$	is the standard uncertainty associated to Xi

$\frac{\partial C_{NO,ppm}}{\partial Xi} u(Xi)$ is the partial uncertainty related to *Xi*

The development of equation (D.2) gives (D.3):

$$\begin{aligned}
u^2(C_{NO,ppm}) = &\left(\frac{\partial C_{NO,ppm}}{\partial C_{NO,read}}\right)^2 \times u^2(C_{NO,read}) + \left(\frac{\partial C_{NO,ppm}}{\partial Corr_{fit}}\right)^2 \times u^2(Corr_{fit}) + \left(\frac{\partial C_{NO,ppm}}{\partial Corr_{0,dr}}\right)^2 \times u^2(Corr_{0,dr}) \\
&+ \left(\frac{\partial C_{NO,ppm}}{\partial Corr_{s,dr}}\right)^2 \times u^2(Corr_{s,dr}) + \left(\frac{\partial C_{NO,ppm}}{\partial Corr_{rep}}\right)^2 \times u^2(Corr_{rep}) + \left(\frac{\partial C_{NO,ppm}}{\partial Corr_{adj}}\right)^2 \times u^2(Corr_{adj}) \\
&+ \left(\frac{\partial C_{NO,ppm}}{\partial Corr_{temp}}\right)^2 \times u^2(Corr_{temp}) + \left(\frac{\partial C_{NO,ppm}}{\partial Corr_{a,press}}\right)^2 \times u^2(Corr_{a,press}) + \left(\frac{\partial C_{NO,ppm}}{\partial Corr_{s,vf}}\right)^2 \times u^2(Corr_{s,vf}) \\
&+ \left(\frac{\partial C_{NO,ppm}}{\partial Corr_{V}}\right)^2 \times u^2(Corr_{V}) + \left(\frac{\partial C_{NO,ppm}}{\partial Corr_{\text{int}}}\right)^2 \times u^2(Corr_{\text{int}})
\end{aligned} \tag{D.3}$$

D.4.1.2 Calculation of the partial uncertainties

— Sensitivity coefficients

$$\begin{aligned}
&\frac{\partial C_{NO,ppm}}{\partial C_{NO,read}} = \frac{\partial C_{NO,ppm}}{\partial Corr_{fit}} = \frac{\partial C_{NO,ppm}}{\partial Corr_{0,dr}} = \frac{\partial C_{NO,ppm}}{\partial Corr_{s,dr}} = \frac{\partial C_{NO,ppm}}{\partial Corr_{rep}} = \frac{\partial C_{NO,ppm}}{\partial Corr_{adj}} \\
&= \frac{\partial C_{NO,ppm}}{\partial Corr_{temp}} = \frac{\partial C_{NO,ppm}}{\partial Corr_{a,press}} = \frac{\partial C_{NO,ppm}}{\partial Corr_{s,vf}} = \frac{\partial C_{NO,ppm}}{\partial Corr_{V}} = \frac{\partial C_{NO,ppm}}{\partial Corr_{\text{int}}} = 1
\end{aligned}$$

— $u(C_{NO,read})$: uncertainty related to the reading of the concentration is due to the resolution of the analyser and of the data acquisition. The uncertainty can be considered as negligible.

— $u(Corr_{fit})$: if $X_{fit,\max}$ is the performance characteristic equal to the maximum deviation between the measured value and the value given through the linear regression achieved during the laboratory test, then, we can consider that the lack of fit has an equal probability to take any value included in the interval [-$X_{fit,max}$; + $X_{fit,max}$]. The standard uncertainty is calculated by application of a rectangular probability distribution.

If the performance characteristic is expressed in % of the range:

$$u(Corr_{fit}) = \frac{X_{fit,\max}/100 \times range}{\sqrt{3}}$$

$$\frac{\partial C_{NO,ppm}}{\partial Corr_{fit}} \times u(Corr_{fit}) = \frac{X_{fit,\max}/100 \times Range}{\sqrt{3}}$$

— $u(Corr_{0,dr})$, $u(Corr_{s,dr})$: where $X_{0,dr}$ and $X_{s,dr}$ are the zero and the span drift, we can suppose that the drifts have the same probability to take any value included in the intervals [-$X_{0,dr}$; + $X_{0,dr}$] and [-$X_{s,dr}$; + $X_{s,dr}$]. The standard uncertainties are calculated by application of a rectangular probability distribution:

$$u(Corr_{0,dr}) = \frac{X_{0,dr}}{\sqrt{3}} \text{ and } u(Corr_{s,dr}) = \frac{X_{s,dr}}{\sqrt{3}}$$

Then $\dfrac{\partial C_{NO,ppm}}{\partial Corr_{0,dr}} \times u(Corr_{0,dr}) = \dfrac{X_{0,dr}}{\sqrt{3}}$ and $\dfrac{\partial C_{NO,ppm}}{\partial Corr_{s,dr}} \times u(Corr_{s,dr}) = \dfrac{X_{s,dr}}{\sqrt{3}}$

— $u(Corr_{rep})$: standard uncertainty due to the repeatability is equal to the repeatability standard deviation calculated from the results of the repetitions of the measurements.

Several tests can be carried out under different concentrations (at least at zero and at one span level). But only one of the values shall be included in the calculation:

— choose the repeatability standard deviation corresponding to the closest concentration measured in stack (in the example: the ELV);

— or the highest repeatability (relative) standard deviation, regardless of the concentration measured in stack;

— in the example, the highest repeatability standard deviation has been chosen:

$$u(Corr_{rep}) = \max(S_{0,rep}\ ; S_{s,rep}) = S_{rep}$$

where

$S_{r,0}$ is the standard uncertainty at level zero

$S_{r,s}$ is the standard uncertainty at span level

Then $\dfrac{\partial C_{NO,ppm}}{\partial Corr_{rep}} \times u(Corr_{rep}) = S_{rep}$

— standard uncertainties associated to the influence quantities $u(Corr_{temp})$, $u(Corr_{a,press})$, $u(Corr_{s,vf})$, $u(Corr_V)$: during the laboratory test, the influence quantities are tested for one value of the parameter and the effects of the influence quantities are supposed to be proportional to the value of the parameter. The correction of the effect of an influence quantity is also proportional to its variation.

If *Xj* is an influence quantity: $Corr_{Xj} = C_{j,\inf} \times \Delta X_j$

where

$C_{j,\inf}$ is a constant,

ΔX_j is the variation of the influence quantity.

The uncertainty associated to the correction is given by:

$$u^2(Corr_{Xj}) = \left(\frac{\partial Corr_{Xj}}{\partial \Delta X_j}\right)^2 \times u^2(\Delta X_j)$$

where

$\dfrac{\partial Corr_{Xj}}{\partial \Delta X_j} = C_{j,\inf}$ is the sensitivity coefficient; the sensitivity factor corresponds to the effect of the variation ΔX_j of the influence quantity in the response of the analyser. Sensitivity coefficients are determined by laboratory tests.

$u(\Delta X_j)$ is the standard uncertainty associated to the variation of the influence quantity between the calibration period and the measuring period. The influence quantity has the same probability to take any value included in the interval ΔX_j.

In the standard EN ISO 14956, the calculation of the standard uncertainty associated with the variation of the influence quantity is given by the equation:

$$u(\Delta Xj) = \sqrt{\frac{(x_{j,\max} - x_{j,adj})^2 + (x_{j,\min} - x_{j,adj}) \times (x_{j,\max} - x_{j,adj}) + (x_{j,\min} - x_{j,adj})^2}{3}}$$

where

$x_{j,\min}$ is the minimum value of the influence quantity X_j during the measuring period

$x_{j,\max}$ is the maximum value of the influence quantity X_j during the measuring period

$x_{j,adj}$ is the value of the influence quantity X_j during the adjustment of the analyser

Then: $\dfrac{\partial C_{NO,ppm}}{\partial Corr_{Xj}} \times u(Corr_{Xj}) = \dfrac{\partial Corr_{Xj}}{\partial X_j} \times u(\Delta X_j)$

$$\frac{\partial C_{NOppm}}{\partial Corr_{Xj}} \times u(Corr_{Xj}) = c_j \sqrt{\frac{(x_{j,\max} - x_{j,adj})^2 + (x_{j,\min} - x_{j,adj}) \times (x_{j,\max} - x_{j,adj}) + (x_{j,\min} - x_{j,adj})^2}{3}}$$

The calculation can be simplified in the two following cases:

— first case: if the value $X_{j,adj}$ is at the centre of the interval $\left[X_{j,\max} ; X_{j,\min}\right]$

$$u(\Delta X_j) = \frac{(x_{j,\max} - x_{j,\min})}{\sqrt{12}} \text{ and } \frac{\partial C_{NO,ppm}}{\partial Corr_{Xj}} \times u(Corr_{Xj}) = c_j \times \frac{(x_{j,\max} - x_{j,\min})}{\sqrt{12}}$$

— second case: if the value $X_{j,adj}$ is equal to $X_{j,\max}$ or $X_{j,\min}$:

$$u(\Delta X_j) = \frac{(x_{j,\max} - x_{j,\min})}{\sqrt{3}} \text{ and } \frac{\partial C_{NO,ppm}}{\partial Corr_{Xj}} \times u(Corr_{Xj}) = c_j \times \frac{(x_{j,\max} - x_{j,\min})}{\sqrt{3}}$$

— standard uncertainties associated with the interferents: the case of the interferents is similar to the case of influence quantities. The effect of the interferents is tested for one concentration of interferent and is likely to be proportional to the value of the parameter. The correction of the effect of an influence quantity is also proportional to its variation.

If Int_j is an interferent for the measured compound, and ΔInt_j its variation interval during the measurement period, the standard uncertainty of the correction is given by:

$$u^2(Corr_{Intj}) = \left(\frac{\partial Corr_{Intj}}{\partial \Delta Int_j} \right)^2 \times u^2(\Delta Int_j)$$

where

$\frac{\partial Corr_{Intj}}{\partial \Delta Int_j}$ is the sensitivity coefficient ; sensitivity coefficients are determined by laboratory tests

$u(\Delta Int_j)$ is the standard uncertainty associated to the variation of the concentration of the interferent Int_j during the measuring period. We can suppose that the influence quantity has the same probability to take any value included in the interval ΔInt_j

In the standard EN ISO 14956, the calculation of the standard uncertainty associated to the variation of the influence quantity is given by the equation:

$$u(\Delta Int_j) = \sqrt{\frac{(Int_{j,\max} - Int_{j,adj})^2 + (Int_{j,\min} - Int_{j,adj}) \times (Int_{j,\max} - Int_{j,adj}) + (Int_{j,\min} - Int_{j,adj})^2}{3}}$$

where

$Int_{j,\min}$ is the minimum concentration of the interferent j during the measuring period

$Int_{j,\max}$ is the maximum concentration of the interferent j during the measuring period

$Int_{j,adj}$ is the concentration of the interferent j in the calibration gas used to adjust the analyser

Then: $\frac{\partial C_{NO,ppm}}{\partial Corr_{Intj}} \times u(Corr_{Intj}) = \left(\frac{\partial Corr_{Intj}}{\partial \Delta Int_j} \right) \times u(\Delta Int_j)$

$$\frac{\partial C_{NO,ppm}}{\partial Corr_{Intj}} \times u(Corr_{Intj}) = \frac{C_j}{Int_{j,test}} \times \sqrt{\frac{(Int_{j,\max} - Int_{j,adj})^2 + (Int_{j,\min} - Int_{j,adj}) \times (Int_{j,\max} - Int_{j,adj}) + (Int_{j,\min} - Int_{j,adj})^2}{3}}$$

where

$C_j = \frac{\partial Corr_{Intj}}{\partial \Delta Int_j}$ is the sensitivity coefficient

$Int_{j,test}$ is the concentration of the interferent used to determine the sensitivity coefficient C_j during laboratory test

The calculation can be simplified if the value $X_{j,adj}$ is equal to zero, that is to say that the concentration of the interferent in the calibration gas is equal to zero, what is often the case. The standard uncertainty is then given by:

$$u(\Delta Int_j) = \sqrt{\frac{Int_{j,\max}{}^2 + Int_{j,\min} \times Int_{j,\max} + Int_{j,\min}{}^2}{3}}$$

$$\frac{\partial C_{NO,ppm}}{\partial Corr_{Intj}} \times u(Corr_{Intj}) = \frac{C_j}{Int_{j,test}} \times \sqrt{\frac{Int_{j,\max}{}^2 + Int_{j,\min} \times Int_{j,\max} + Int_{j,\min}{}^2}{3}}$$

All interferents shall not be included in the calculation of the global uncertainty. In accordance with the standard EN ISO 14956, the calculation shall include the highest value between the sum $S_{\text{int,p}}$ of the interferents with positive impact in the response of the analyser and the sum $S_{\text{int,n}}$ of the interferents with negative impact.

$$S_{\text{int,p}} = \frac{\partial C_{NO,ppm}}{\partial Corr_{Int,p1}} \times u(Corr_{\text{int,p1}}) + \frac{\partial C_{NO,ppm}}{\partial Corr_{Int,p2}} \times u(Corr_{\text{int,p2}}) + \ldots\ldots + \frac{\partial C_{NO,ppm}}{\partial Corr_{Int,pk}} \times u(Corr_{\text{int,pk}})$$

$$S_{\text{int},n} = \frac{\partial C_{NO,ppm}}{\partial Corr_{Int,n1}} \times u(Corr_{\text{int,n1}}) + \frac{\partial C_{NO,ppm}}{\partial Corr_{Int,n2}} \times u(Corr_{\text{int,n2}}) + \ldots\ldots + \frac{\partial C_{NO,ppm}}{\partial Corr_{Int,nt}} \times u(Corr_{\text{int,nt}})$$

where

$S_{\text{int},p}$ is the sum of the interferents with positive impact

$\frac{\partial C_{NO,ppm}}{\partial Corr_{Int,p1}} \times u(Corr_{\text{int,p1}})$ is the partial uncertainty due to the 1st interferent with positive impact

$\frac{\partial C_{NO,ppm}}{\partial Corr_{Int,pk}} \times u(Corr_{\text{int,pk}})$ is the partial uncertainty due to the kth interferent with positive impact

$S_{\text{int},n}$ is the sum of the interferents with negative impact

$\frac{\partial C_{NO,ppm}}{\partial Corr_{Int,n1}} \times u(Corr_{\text{int,n1}})$ is the partial uncertainty due to the 1st interferent with negative impact

$\frac{\partial C_{NO,ppm}}{\partial Corr_{Int,nt}} \times u(Corr_{\text{int,nt}})$ is the partial uncertainty due to the t[th] interferent with negative impact

$$\left(\frac{\partial C_{NO,ppm}}{\partial Corr_{\text{int}}}\right)^2 \times u^2(Corr_{\text{int}}) = \max\left[S_{\text{int,p}}{}^2; S_{\text{int,n}}{}^2\right]$$

— $u(Corr_{adj})$: standard uncertainty $u(Corr_{adj})$ is calculated from the uncertainty of the calibration gas. In general the uncertainty given by the manufacturer is the expanded uncertainty; if X_{cal} is the expanded uncertainty of the calibration gas expressed in % relative, the standard uncertainty of the correction at ELV level is:

$$u(Corr_{adj}) = \frac{X_{cal}/100 \times ELV}{2}$$

Then: $\frac{\partial C_{NO,ppm}}{\partial Corr_{adj}} \times u(Corr_{adj}) = \frac{X_{cal}/100 \times ELV}{2}$

D.4.2 NOx measurement

NOx concentration and standard uncertainty expressed in ppm are given by:

$$C_{NOx,ppm} = C_{NO,ppm} + C_{NO2,ppm}$$

$$u^2(C_{NOx,ppm}) = u^2(C_{NO,ppm}) + u^2(C_{NO2,ppm})$$

If the converter efficiency is not equal to 100 %, the NO$_x$ concentration $C_{NOx,stack}$ in the stack is not equal to the NO$_x$ concentration $C_{NOx,ppm}$ given by the analyser. The $C_{NOx,stack}$ is equal to the sum of the NO_2 concentration $C_{NO2,stack}$ and the NO concentration $C_{NO,stack}$ in stack. For the NO measurement, the converter has any effect and it can be considered that $C_{NO,stack} = C_{NO,ppm}$, *where* $C_{NO,ppm}$ is the concentration given by the analyser.

$$C_{NOx,stack} = C_{NO,stack} + C_{NO2,stack}$$

$$C_{NOx,stack} = C_{NO,ppm} + \frac{C_{NO2,ppm}}{\eta} \times 100$$

$$C_{NOx,stack} = C_{NO,ppm} + \frac{C_{NOx,ppm} - C_{NO,ppm}}{\eta} \times 100$$

The values $C_{NO,ppm}$ and $C_{NOx,ppm}$ are correlated.

Then the standard uncertainty associated to the NOx concentration is given by:

$$u_c^2(C_{NOx,stack}) = \left(\frac{\partial C_{NOx,stack}}{\partial C_{NO,ppm}}\right)^2 \times u^2(C_{NO,ppm}) + \left(\frac{\partial C_{NOx,stack}}{\partial C_{NOx,ppm}}\right)^2 \times u^2(C_{NOx,ppm})$$

$$+\left(\frac{\partial C_{NOx,stack}}{\partial \eta}\right)^2 \times u^2(\eta)$$

$$+2\times\left(\frac{\partial C_{NOx,stack}}{\partial C_{NO,ppm}}\right)\times\left(\frac{\partial C_{NOx,stack}}{\partial C_{NOx,ppm}}\right)\times u(C_{NO,ppm})\times u(C_{NOx,ppm})\times \mathrm{r}(C_{NO,ppm},C_{NOx,ppm})$$

where

$$\frac{\partial C_{NOx,stack}}{\partial C_{NO,ppm}} = \frac{\eta - 100}{\eta}$$

$$\frac{\partial C_{NOx,stack}}{\partial C_{NOx,ppm}} = \frac{100}{\eta}$$

$$\frac{\partial C_{NOx,stack}}{\partial \eta} = \frac{(-C_{NOx,ppm} + C_{NO,ppm})\times 100}{\eta^2}$$

$\mathrm{r}(C_{NO,ppm},C_{NOx,ppm})$: correlation factor.

$$u_c^2(C_{NOx,stack}) = \left(\frac{\eta-100}{\eta}\right)^2 \times u^2(C_{NO,ppm}) + \left(\frac{100}{\eta}\right)^2 \times u^2(C_{NOx,ppm})$$

$$+\left(\frac{100\times(C_{NO,ppm} - C_{NOx,ppm})}{\eta^2}\right)^2 \times u^2(\eta)$$

$$+2\times\left(\frac{\eta-100}{\eta}\right)\times\left(\frac{100}{\eta}\right)\times u(C_{NO,ppm})\times u(C_{NOx,ppm})\times \mathrm{r}(C_{NO,ppm},C_{NOx,ppm})$$

Where $u(C_{NOx,ppm})$ is calculated in the same way as $u(C_{NO,ppm})$.

As the variation in the NOx concentration is in the same direction as the variation of the NO concentration: $\mathrm{r}(C_{NO,ppm},C_{NOx,ppm})$ is positive or equal to zero. The value $\left(\eta - 100/\eta\right)\left(100/\eta\right)$ is negative; then the last term of the equation is negative. The uncertainty is maximised by considering this term as equal to zero.

In this case the calculation is simplified and given by:

$$u_c^2(C_{NOx,stack}) = \left(\frac{\eta - 100}{\eta}\right)^2 \times u^2(C_{NO,ppm}) + \left(\frac{100}{\eta}\right)^2 \times u^2(C_{NOx,ppm})$$

$$+ \left(\frac{100 \times (C_{NO,ppm} - C_{NOx,ppm})}{\eta^2}\right)^2 \times u^2(\eta)$$

D.4.3 Results of standard uncertainties calculation

Studied concentration of NO_x : ELV for the site = 200 mg/m^3 eq NO_2

$C_{\text{NOx,stack}}$ = 200 mg/m^3 eq. NO_2 = 97,39 ppm

With converter efficiency equal to 98 %, the NO_x concentration measured by the analyser is equal to:

$C_{\text{NOx,ppm}}$ = 97,27 ppm.

$C_{\text{NO,stack}}$ = $C_{\text{NO,ppm}}$ =94% of C_{NOx} = 91,55 ppm

$C_{\text{NO2,ppm}}$ = 5,73 ppm

$C_{\text{NO2,stack}}$= 5,84 ppm

Table D.3 — Results of uncertainty calculation

Performance characteristics	Partial standard uncertainty	Value of partial standard uncertainty at limit value (in ppm)
Lack of fit	$u(Corr_{fit})$	$\frac{(0,7/100)\times 200}{\sqrt{3}} = 0,81$
Zero drift	$u(Corr_{0,dr})$	$\frac{(0,01/100)\times 200}{\sqrt{3}} = 0,01$
Span drift	$u(Corr_{s,dr})$	$\frac{(1/100)\times 200}{\sqrt{3}} = 1,16$
Repeatability in laboratory at span level	$u(Corr_{rep})$	$(0,8/100) \text{ x } 200 = 1,60$
Sensitivity to the sample volume flow	$u(Corr_{s,vf})$	$\frac{(1/100)}{\sqrt{3}}\times 200\times\frac{10}{\sqrt{3}} = 1,16$
Sensitivity to atmospheric pressure	$u(Corr_{apress})$	$(0,8/100)\times 97,3\frac{100-99}{\sqrt{3}} = 0,45$ for NOx $(0,8/100)\times 91,6\frac{100-99}{\sqrt{3}} = 0,43$ for NO
Sensitivity to ambient temperature	$u(Corr_{temp})$	$\frac{(1/100)}{10} \text{ x } 200 \text{ x } \sqrt{\frac{(308-285)^2+(308-285)(283-285)+(283-285)^2}{3}}$ $= 2,549$
Sensitivity to electric voltage	$u(Corr_{volt})$	$\frac{(0,12/100)}{10} \text{ x } 200 \frac{(230+230/100\times 5)-(230-230/100\times 5)}{\sqrt{12}}$ $= 0,16$
Interferent: NH_3	$u(Corr_{NH3})$	$\frac{0,75}{20} \text{ x } \sqrt{\frac{20^2}{3}} = 0,44$
Interferent: CO_2	$u(Corr_{CO2})$	$\frac{-2,6}{15} \text{ x } \sqrt{\frac{15^2+15\times 8+8^2}{3}} = -2,03$
Uncertainty of calibration gas	$u(Corr_{adj})$	$\frac{(2/100) \text{ x } 97,27}{2} = 0,98$ for NOx $\frac{(2/100) \text{ x } 91,55}{2} = 0,92$ for NO $\frac{(2/100) \text{ x } 5,7}{2} = 0,057$ for NO_2
Converter efficiency	$u(\eta)$	$\sqrt{\left(\frac{3}{\sqrt{3}}\right)^2 + 1,0^2} = 2,00\ \%$ % (absolute)

D.4.4 Calculation of combined uncertainties

D.4.4.1 NO measurement

— Combined uncertainty:

$$u(Corr_{rep}) = \max(S_{0,rep}\,; S_{s,rep}) = S_{s,rep} \qquad \text{and} \qquad S_{\text{int,n}} > S_{\text{int,p}}$$

$$u(C_{NO,stack}) = u(C_{NO,ppm}) = \sqrt{\begin{aligned} & u^2(Corr_{fit}) + u^2(Corr_{0,dr}) + u^2(Corr_{s,dr}) + u^2(Corr_{rep}) + u^2(Corr_{s,vf}) \\ & + u^2(Corr_{a,press,NO}) + u^2(Corr_{temp}) + u^2(Corr_{volt}) + u^2(Corr_{CO2}) + u^2(Corr_{adj,NO}) \end{aligned}}$$

$$u(C_{NO,ppm}) = 4{,}1 \text{ ppm}$$

D.4.4.2 NOx measurement

— Combined uncertainty:

$$u(C_{NOx,ppm}) = \sqrt{\begin{aligned} & u^2(Corr_{fit}) + u^2(Corr_{0,dr}) + u^2(Corr_{s,dr}) + u^2(Corr_{rep}) + u^2(Corr_{s,vf}) \\ & + u^2(Corr_{a,press,NOx}) + u^2(Corr_{temp}) + u^2(Corr_{volt}) + u^2(Corr_{CO2}) + u^2(Corr_{adj,NOx}) \end{aligned}}$$

$$u(C_{NOx,ppm}) = 4{,}1 \text{ ppm}$$

$$u_c^{\,2}(C_{NOx,stack}) = \left(\frac{\eta - 100}{\eta}\right)^2 \times u^2(C_{NO,ppm}) + \left(\frac{100}{\eta}\right)^2 \times u^2(C_{NOx,ppm})$$

$$+ \left(\frac{100 \times (C_{NO,ppm} - C_{NOx,ppm})}{\eta^2}\right)^2 \times u^2(\eta)$$

$$u(C_{NOx,stack}) = 4{,}2 \text{ ppm}$$

D.5 Conversion of the concentrations in mg/m^3

The concentration in mg/m^3 is calculated according to the equation:

$$C_{mg/m3} = C_{ppm} \times \frac{M_{mol}}{V_{mol,std}}$$

If the uncertainties associated with M_{mol} and $V_{mol,std}$ are neglected (uncertainties are due to the fact that values are rounded and depend of the number of digits), the standard uncertainty associated with the result calculated in mg/m^3 is given by:

$$u^2(C_{mg/m3}) = \left(\frac{M_{mol}}{V_{mol,std}}\right)^2 \times u^2(C_{ppm})$$

D.5.1 No measurement

$C_{NO,mg/m3} = 122{,}6$ mg/m^3

— Combined uncertainty :

$u(C_{NO,mg/m3}) = 5{,}4$ mg/m^3

— Overall uncertainty :

$U(C_{NO,mg/m3}) = \pm 10{,}8$ mg/m^3 (k = 2)

$U(C_{NO,mg/m3,rel}) = \pm 8{,}8$ % relative (k=2)

D.5.2 NO_2 measurement

$C_{NO2,mg/m3} = 12{,}0$ mg/m^3

D.5.3 NO_x measurement

$C_{NO2,mg/m3} = 200{,}0$ mg/m^3 equivalent NO_2

D.5.4 Combined uncertainty

$u(C_{NOx,mg/m3}) = 8{,}5$ mg/m^3 eq. NO_2

D.5.5 Overall uncertainty

$U(C_{NOx,mg/m3}) = \pm 17{,}0$ mg/m^3 eq. NO_2 (k = 2)

$U(C_{NOx,mg/m3,rel}) = \pm 8{,}5$ % relative (k = 2)

D.6 Evaluation of the compliance with the required measurement quality

Interferents with a positive impact:

$Total = \frac{0{,}75}{200} \times 100 = 0{,}37\ \%$ of the range Total < 4% of the range

Interferents with a negative impact:

$Total = \frac{2{,}6}{200} \times 100 = 1{,}3\ \%$ of the range Total < 4% of the range

All values of the performance characteristics obtained in tests are complying with the requirements.

Conclusion: the measurement method fulfils the requirements.

B C D E F G H

2 **Adjustment and span check after measurement**

3

	Gas	expected value	reading before	reading after
4	**Gas**	**expected value**	**reading before**	**reading after**
5	Span	918,9	912,3	882,4
6	zero	0	1,8	1,7
7	Time		5,36	17,58
8	duration		F7-E7	
9	In minutes		742	

Fill in only boxes with shaded area

10

11 **Relation between the true value (Y) and the reading (X), Y=AX+B**

12

	Coefficients	Adjustment	Check	deviation	Drift / min.
13	**Coefficients**	**Adjustment**	**Check**	**deviation**	**Drift / min.**
14	A (span)	(D5-D6)/(E5-E6)	(D5-D6)/(F5-F6)	E13-D13	(E13-D13)/E9
15	B(zero)	D13*-E6	E13*-F6	E14-D14	(E14-D14)E9

16

17 **Drift is 1-((D5-(D5*F13+F14))/D5)**

Equation to calculate the corrected concentration (Ccor.) from the reading (C) and time (t)

18

19 C cor. = C x (C14 + E14 x t) + (C15 + E15 x t)

20

21 **For this application :**

22 *CONCATENER("C cor. = C x (";CTXT(D13;4);" + ";CTXT(G13;4);" x t) + (";CTXT(D14;4);" + ";CTXT(G14;4);" x t)")*

23 **Example**

24

25

	C	Time	C cor.
26	**C**	**Time**	**C cor.**
27–28	E5	0	D28*(D$13+G$13*E28)+(D$14+G$14*E28)
29	F5	E9	D29*(D$13+G$13*E29)+(D$14+G$14*E29)
30	180	124	D30*(D$13+G$13*E30)+(D$14+G$14*E30)

31

Annex E
(informative)

Procedure of correction of data from drift effect

The drift is most likely to be proportional to time.

B C D E F G H

2 **Adjustment and span check after measurement**

3

	Gas	expected value	reading before	reading after
4	Gas	expected value	reading before	reading after
5	Span	918,9	912,3	882,4
6	zero	0	1,8	1,7
7	Time		5,36	17,58
8	duration		F7-E7	
9	In minutes		742	

10

Fill in only boxes with shaded area

11 **Relation between the true value (Y) and the reading (X), Y=AX+B**

12

	Coefficients	Adjustment	Check	deviation	Drift / min.
13	Coefficients	Adjustment	Check	deviation	Drift / min.
14	A (span)	(D5-D6)/(E5-E6)	(D5-D6)/(F5-F6)	E13-D13	(E13-D13)/E9
15	B(zero)	D13*-E6	E13*-F6	E14-D14	(E14-D14)E9

16

17 **Drift is 1-((D5-(D5*F13+F14))/D5)**

Equation to calculate the corrected concentration (Ccor.) from the reading (C) and time (t)

18

19 C cor. = C x (C14 + E14 x t) + (C15 + E15 x t)

20

21 **For this application :**

22 *CONCATENER("C cor. = C x (";CTXT(D13;4);" + ";CTXT(G13;4);" x t) + (";CTXT(D14;4);" + ";CTXT(G14;4);" x t)")*

23 **Example**

24

25

	C	Time	C cor.
26	C	Time	C cor.
27			
28	E5	0	D28*(D$13+G$13*E28)+(D$14+G$14*E28)
29	F5	E9	D29*(D$13+G$13*E29)+(D$14+G$14*E29)
30	180	124	D30*(D$13+G$13*E30)+(D$14+G$14*E30)

31

Annex F
(informative)

Evaluation of the method in the field

F.1 General

The method has been evaluated during six field tests, on waste incineration installations, co-incineration installations and large combustion plants. Each test was performed by at least four different European measuring teams originating from ten member countries.

F.2 Characteristics of installations

— 1st field test: INERIS bench-loop at Verneuil en Halatte (F); the bench-loop simulates combustion or waste incineration exhaust gases. Five teams took part in the 1st field test. Double measurements were not performed simultaneously but sequentially. Five different flue gas matrices were generated. Within each matrix, two sequential measurements were performed. Two additional sequential measurements were performed in flue gas matrices where the flue gas concentrations varied. There were a total of twelve measurements performed by all the teams.

— 2nd field test: waste incinerator in Denmark. Four teams took part to the field test and performed double measurements simultaneously. A total of sixteen measurements were performed by all the teams.

— 3rd field test: waste incinerator in Italy. Four teams took part to the field test. Two pairs of two teams performed double measurements simultaneously and the four teams performed single measurements simultaneously. A total of six double measurements were performed by each pair of two teams while a total of twelve single measurements was performed by all teams.

— 4th field test: co-incinerator combined heat and power installation in Sweden. The fluidised bed boilers operate on fuel mixes of wood chips, demolition waste, peat and coal. Two pairs of two teams performed double measurements simultaneously and the four teams performed single measurements simultaneously. A total of six double measurements were performed by each pair of two teams while a total of twelve single measurements was performed by all the teams.

— 5th field test: co-incinerator cement plant in Germany. The fuel could be coal, heavy oil and secondary fuel (e.g. paper, plastics, textiles, and tires). Four teams took part to the field test and performed doubles measurements simultaneously. All the teams performed a total of sixteen double measurements.

— 6th field test: coal fired power plant in Germany. Four teams performed their double measurements simultaneously. The total amount of double measurements performed by all teams was 12.

An overview of the flue gas characteristics is given in Table F.1.

Table F.1 — Flue gas characteristics during field tests

Field test	Installation	Fuel	Flue gas characteristics						
			T °C	O_2 % vol	NOx mg/m^3	SO_2 mg/m^3	CO mg/m^3	H_2O % vol	PM mg/m^3
1	Power plant [a]	Natural gas	< 150	5 to 13	10 to 1300	10 to 2000	20 to 400	10 to 21	<1
2	Waste incinerator	Municipal waste	90 to 110	8 to 11	180 to 250	25 to 250	5 to 15	13 to 19	1 to 5
3	Waste incinerator	Municipal waste	85 to 105	16 to 18	140 to 210	5 to 50	0 to 2	8 to 12	1 to 5
4	Co-incinerator	Wood, waste peat, coal	70 to 80	4 to 6	4 to 70	0 to 10	50 to 150	8 to 12	0 to 20
5	Co-incinerator	Coal, oil, waste	140 to 170	4 to 6	440 to 1060	60 to 170	260 to 740	23 to 26	5 to 10
6	Power plant	Coal	130 to 140	8,9 to 9,2	110 to 140	1000 to 1130	3 to 6	5,5 to 8	< 50

[a] bench-loop: flue gas simulation

F.3 Repeatability and reproducibility in the field

F.3.1 General

Repeatability standard deviation s_r and reproducibility standard deviation s_R are determined by performing inter-laboratory tests.

Repeatability standard deviation s_r, internal confidence interval (CI_r) and repeatability in the field r are calculated according to ISO 5725-2 and ISO 5725-6, from the results of the double measurements implemented by the same laboratory.

$$CI_r = t_{0,95;n-1} \times s_r \qquad \text{and} \qquad r = \sqrt{2} \times t_{0,95;n-1} \times s_r$$

where

CI_r is the internal confidence interval

s_r is the repeatability standard deviation

$t_{0,95;n-1}$ is the student factor for a level of confidence of 95% and a degree of freedom of n-1 (n: number of double measurements)

r is the repeatability in the field

Reproducibility standard deviation s_R, external confidence interval (CI_R) and reproducibility in the field R are calculated according to ISO 5725-2, from the results of parallel measurements performed simultaneously by several laboratories.

$$CI_R = t_{0,95;np-1} \times s_R \qquad \text{and} \qquad R = \sqrt{2} \times t_{0,95;np-1} \times s_R$$

where

CI_R is the external confidence interval

s_R is the repeatability standard deviation

$t_{0,95;np-1}$ is the student factor for a level of confidence of 95% and a degree of freedom of np-1 (n: number of measurements; p: number of laboratories)

R is the reproducibility in the field

F.3.2 Repeatability

Table F.2.1 — Repeatability in the field

Field test	**Concentration**		**Number of teams**	**Number of double measurements**	**Repeatability standard deviation s_r**	**95% confidence interval for a single measurement CI_r**	**Repeatability r**	
	Range mg/m^3	Average mg/m^3					mg/m^3	%
1A		30	5	2	0,44	1,0	1,4	4,8
1B		50	5	2	0,59	1,3	1,9	3,8
1C		90	5	2	0,83	1,9	2,7	3,0
1D		450	5	2	3,3	7,5	10,6	2,3
1E		1350	5	2	9,4	21	30,1	2,2
1F	600 to 650	625	5	1	4,4	12	17,3	2,8
1G	87 to 98	92	5	1	0,88	2,4	3,5	3,8
2	187 to 243	217	4	16	3,6	7,6	10,7	5,0
3	61 to 78	68	4	12	2,4	5,3	7,4	10,3
4	4,5 to 67	35	4	12	0,61	1,4	2,0	7,3
5	448 to 1065	634	3	12	9,2	20	28,3	4,5
6	118 to 137	126	4	12	2,3	5,0	7,1	5,6

The following functions were determined:

$s_r(C) = 0,0133\ C + 0,757$ in mg/m^3

$s_{r\ limit}(C) = 0,0153\ C + 2,1$ in mg/m^3

$CI_r(C) = 0,029\ C + 2$ in mg/m^3

$r(C) = 0,041\ C + 2,8$ in mg/m^3

F.3.3 Reproducibility

Table F.2.2 — Reproducibility in the field

Field test	Concentration		Number of teams	Number of measurements	Reproducibility standard deviation s_R	95% confidence interval for a single measurement CI_R	Reproducibility R	
	Range mg/m^3	Average mg/m^3					mg/m^3	%
1A		30	5	2	0,59	1.3	1,9	6,4
1B		50	5	2	0,92	2.1	3,0	5,9
1C		90	5	2	1,0	2.3	3,2	3,6
1D		450	5	2	3,1	7.0	9,9	2,2
1E		1350	5	2	19	44	61,6	4,6
1F	600 to 650	625	5	1	22	60	84,6	13,5
1G	87 to 98	92	5	1	4,0	11	15,9	17,2
2	187 to 243	217	4	16	4,6	9.7	13,8	6,3
3	61 to 78	68	4	12	4,6	10	14,2	21
4	4,5 to 67	35	4	12	0,039 [NOx]+0,52	4	5,8	16,6
5	448 to 1065	634	3	12	12,0	27	38,2	6,0
6	118 to 137	126	4	12	4,1	9,0	12,7	10,1

The following functions were determined:

$S_R(C) = 0{,}0153\ C + 2{,}12$ in mg/m^3

$CI_R(C) = 0{,}038\ C + 4{,}4$ in mg/m^3

$R(C) = 0{,}054\ C + 6$ in mg/m^3

Annex ZA
(informative)

Relationship with EU Directives

This European Standard has been prepared under a mandate given to CEN by the European Commission and the European Free Trade Association and supports essential requirements of the EU Council Directive 2000/76/EC on the Incineration of Hazardous Waste and of the EU Council Directive 2001/80/EC on the limitation of emissions of certain pollutants into the air from large combustion plants.

WARNING — Other requirements and EU Directives may be applicable falling within the scope of use of this European Standard.

Bibliography

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